

# Sarath Satya Mutyala

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## SUMMARY

SRE with 3 years of experience operating IoT-scale Kubernetes platforms ingesting real-time telemetry from 2M+ connected vehicles via MQTT, Kafka, and ClickHouse across AWS and Azure. I specialize in reliability engineering for stateful, high-throughput systems: SLO-driven observability, GitOps delivery (Argo CD + Helm), and Terraform-based platform automation that supports 30+ engineers and 250+ services at 99.9% availability.

## SKILLS

**Cloud Platforms:** AWS (EKS, EC2, ECR, VPC, S3, Route 53, ELB/ALB, IAM), Azure (AKS, VM, ACR, VNet, Storage, DNS, Load Balancer, Key Vault, RBAC), Linux (RHEL, Amazon Linux 2, Ubuntu)

**IaC & GitOps for Automation:** Terraform, Ansible, Argo CD, Scripting (Python, Bash/Shell)

**CI/CD & Release Management:** Azure DevOps, Jenkins, Git branching, Nexus, blue-green/rolling deployments, SonarQube (Quality gates)

**Containerization & Orchestration:** Docker, Kubernetes, Helm, Kustomize, Cert-Manager

**Security & Compliance:** Secrets Management (Azure Key Vault, AWS Secrets Manager), IAM, RBAC, Security Groups, Kubernetes security controls (RBAC, pod security, network policies)

**Observability, Reliability & Cost Optimization:** Prometheus, Grafana, Loki, ELK/EFK, OpenTelemetry, New Relic; SLO/SLI design, error budgets, on-call alerting, incident response, postmortems; Kubernetes autoscaling & right-sizing via PerfectScale and KEDA (event-driven); AWS Cost Explorer, Azure Cost Management

**Distributed Systems & Databases:** Nginx (Reverse proxy, load balancing), MQTT (EMQX/VerneMQ), Flink (Stream processing), Kafka (Event streaming and messaging), Redis/Dragonfly (Caching), ClickHouse (High-performance analytics database), PostgreSQL, MySQL

## EXPERIENCE

### Site Reliability Engineer | TVS Motor Company Limited | Bengaluru, India | August 2023 - Present

- Operate high-availability telemetry ingestion platform on AWS/Azure Kubernetes that processes real-time MQTT streams from 2M+ connected vehicles into Kafka and ClickHouse, sustaining sub-second analytics query latency at peak ingest.
- Own production reliability for the platform supporting 30+ internal engineers and 250+ services, holding 99.9% availability through multi-region, multi-AZ failover, automated load balancing, and capacity planning.
- Improved Kubernetes cluster efficiency by 30% (CPU/memory utilization) across 250+ workloads — Nginx, EMQX, Kafka, ClickHouse, Redis, Dragonfly — by right-sizing requests/limits, tuning HPA, and resolving recurring throttling and OOM root causes.
- Automated TLS certificate issuance and rotation across AKS clusters using Cert-Manager integrated with Azure Key Vault, eliminating manual renewals and cutting cert-related production incidents from 8 in the prior year to 0 over the following 12 months.
- Built end-to-end observability across 250+ services using Prometheus, Grafana, Loki, EFK, and OpenTelemetry — including SLO/SLI dashboards and burn-rate alerting for MQTT brokers, Kafka, ClickHouse, Redis, and NGINX — reducing MTTR by ~50% and customer-reported incidents by ~90% while sustaining 99.9% monthly SLA.
- Designed multi-stage CI/CD pipelines in Jenkins and Azure DevOps with blue-green and rolling strategies, reducing average release cycle time from 60 min to under 15 min across 250+ microservices.
- Migrated deployment delivery to GitOps using Argo CD with Helm and Kustomize, reaching 60+ production releases/month at 99.5% success rate with automated rollback, and integrated SonarQube quality gates and Azure Key Vault secrets that cut post-release defect rate by 30%.
- Built a library of reusable Terraform modules (VNet/VPC, subnets, NSGs, DNS, load balancers, AKS/EKS) adopted by 7+ teams, cutting new-environment provisioning from 3 days to 2 hours and standardizing network/security baselines across AWS and Azure.
- Cut Kubernetes infrastructure spend by ~40% (~₹8 lakh / ~\$10K per month, sustained 12+ months) across AWS EKS and Azure AKS through KEDA event-driven autoscaling, PerfectScale right-sizing, and node-pool consolidation — savings held without p99 latency SLO regression.
- Tuned high-throughput stateful workloads — ClickHouse query plans, Redis/Dragonfly connection pooling, PostgreSQL replication — to keep p99 read latency under 300 ms at 1000 QPS sustained.
- Primary on-call for 10 critical services on a 1-week rotation across 4 engineers; led incident command for 3 P0 and 20+ P1 incidents, authored 15 postmortems, and drove follow-up actions (runbook automation, alert tuning, recurring root-cause fixes) that reduced repeat-incident rate by ~50%.

## EDUCATION

**Bachelor of Technology in Computer Science | SASTRA Deemed University | Tamil Nadu | 2019 – 2023 | CGPA 8.08/10**